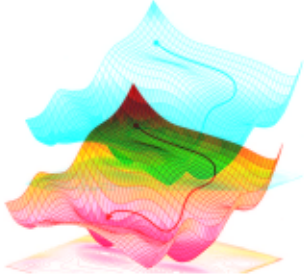
Advanced Risk Minimization

Risk Minimizers



Learning goals

- Know the concepts of the Bayes optimal model (also: risk minimizer, population minimizer)
- Bayes risk
- Consistent learners
- Bayes regret, estimation and approximation error
- Optimal constant model

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- Consistent learners

EMPIRICAL RISK MINIMIZATION

Very often, in ML, we minimize the empirical risk

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

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where

- each observation $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \in \mathcal{X} \times \mathcal{Y}$, so from feature and target space
- $f_{\mathcal{H}} : \mathcal{X} \to \mathbb{R}^g$, f is a model from hypothesis space $\mathcal{H}$; maps a feature vector to output score; sometimes or often we omit $\mathcal{H}$ in the index
- $L : (\mathcal{Y} \times \mathbb{R}^g) \to \mathbb{R}$ is loss;
- $L(y, f)$ measures distance between label and prediction
- We assume that $(\mathbf{x}, y) \sim \mathbb{P}_{xy}$ and $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{xy}$
- $\mathbb{P}_{xy}$ is the distribution of the data generating process (DGP)

Let's define (and minimize) loss in expectation, the theoretical risk:

- and we assume that $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{xy}$ where $\mathbb{P}_{xy}$ is the distribution of the data generating process (DGP).

$$\mathcal{R}(f) := \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] = \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}$$

What is the theoretical justification for this procedure?

TWO SHORT EXAMPLES

Regression with linear model:

- Model: $f(\mathbf{x}) = \boldsymbol{\theta}^\top \mathbf{x} + \theta_0$
- Squared loss: $L(y, f) = (y - f)^2$
- Hypothesis space:

$$\mathcal{H}_{\text{lin}} = \left\{ \mathbf{x} \mapsto \boldsymbol{\theta}^\top \mathbf{x} + \theta_0 : \boldsymbol{\theta} \in \mathbb{R}^d, \theta_0 \in \mathbb{R} \right\}$$

Binary classification with shallow MLP:

- Model: $f(\mathbf{x}) = \pi(\mathbf{x}) = \sigma(\boldsymbol{w}_2^\top \text{ReLU}(\boldsymbol{W}_1 \mathbf{x} + \boldsymbol{b}_1) + b_2)$
- Binary cross-entropy loss:
  $L(y, \pi) = -(y \log(\pi) + (1 - y) \log(1 - \pi))$
- Hypothesis space:

$$\mathcal{H}_{\text{MLP}} = \{ \mathbf{x} \mapsto \sigma(\boldsymbol{w}_2^\top \text{ReLU}(\boldsymbol{W}_1 \mathbf{x} + \boldsymbol{b}_1) + b_2) : \mathbf{W}_1 \in \mathbb{R}^{h \times d}, \mathbf{b}_1 \in \mathbb{R}^h, \mathbf{w}_2 \in \mathbb{R}^h, b_2 \in \mathbb{R} \}$$

OPTIMAL CONSTANTS FOR A LOSS

- Let's assume some RV $z \in \mathcal{Y}$ for a label
  - z not RV y, because we want to fiddle with its distribution
  - Assume z has distribution Q, so $z \sim Q$
  - We can now consider $\arg\min_c \mathbb{E}_{z\sim Q}[L(z,c)]$, so the score-constant which loss-minimally approximates z

We will consider 3 cases for Q

- $Q = P_y$, simply our labels and their marginal distribution in $\mathbb{P}_{xy}$
- $Q = P_{y|x=x}$, conditional label distribution at point $x = \tilde{x}$
- $Q = P_n$, the empirical product distribution for data $y_1, \ldots, y_n$

If we can solve $\arg\min_c \mathbb{E}_{z\sim Q}[L(z,c)]$ for any Q, we will get multiple useful results!

OPTIMAL CONSTANT MODEL

- We would like a loss optimal, constant baseline predictor
- A "featureless" ML model, which always predicts the same value
- Can use it as baseline in experiments, if we don't beat this with more complex model, that model is useless
- Will also be useful as component in algorithms and derivations

OPTIMAL CONSTANT MODEL

- Let's start with the simplest case, L2 loss
- And we want to find and optimal constant model for

$$\arg\min \mathbb{E}[L(z, c)] =$$

$$\arg\min \mathbb{E}[(z - c)^2] =$$

$$\arg\min \mathbb{E}[z^2] - 2cE[z] + c^2 =$$

$$\underset{E[z]}{}$$

- Using $Q = P_y$, this means that, given we know the label distribution, the best constant is $c = E[y]$.
- If we only have data $y_1, \ldots y_n$

$$\arg\min \mathbb{E}_{z \sim P_n}[(z - c)^2] = \mathbb{E}_{z \sim P_n}[z] = \frac{1}{n} \sum_{i=1}^{n} y^{(i)} = \bar{y}$$

- And we want to find and optimal constant model for

RISK MINIMIZER AND APPROXIMATION ERROR

Goal of learning: Train a model $\hat{f}$ for which the true risk $\mathcal{R}_L\left(\hat{f}\right)$ is close to the Bayes risk $\mathcal{R}_L^*$. In other words, we want the **Bayes regret**

$$\mathcal{R}_L\left(\hat{f}\right) - \mathcal{R}_L^*$$

to be as low as possible.

The Bayes regret can be decomposed as follows:

$$\mathcal{R}_L\left(\hat{f}\right) - \mathcal{R}_L^* = \underbrace{\left[\mathcal{R}_L\left(\hat{f}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}_L(f)\right]}_{\text{estimation error}} + \underbrace{\left[\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*\right]}_{\text{approximation error}}$$

Let us assume we are in an "ideal world":

- The hypothesis space $\mathcal{H} = \mathcal{H}_{all}$ is unrestricted. We can choose any measurable $f : \mathcal{X} \to \mathbb{R}$.
- We also assume an ideal optimizer; the risk minimization can always be solved perfectly and efficiently.
- We know $\mathbb{P}_{xy}$.

How should f be chosen?

RISK MINIMIZER AND APPROXIMATION ERROR / 2

The f with minimal risk across all (measurable) functions is called the **risk minimizer**, **population minimizer** or **Bayes optimal model**.

$$
\begin{aligned}
f^*_{\mathcal{H}_{all}} &= \arg\min_{f \in \mathcal{H}_{all}} \mathcal{R}(f) = \arg\min_{f \in \mathcal{H}_{all}} \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] \\
&= \arg\min_{f \in \mathcal{H}_{all}} \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}.
\end{aligned}
$$

The resulting risk is called **Bayes risk**: $\mathcal{R}^* = \mathcal{R}(f^*_{\mathcal{H}_{all}})$

- $\mathcal{R}_L\left(\hat{f}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}(f)$ is the **estimation error**. We fit $\hat{f}$ via empirical risk minimization and (usually) use approximate optimization, so we usually do not find the optimal $f \in \mathcal{H}$.
- $\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}^*_L$ is the **approximation error**. We need to restrict to a hypothesis space $\mathcal{H}$ which might not even contain the Bayes optimal model.

Note that if we leave out the hypothesis space in the subscript it becomes clear from the context!

Similarly, we define the risk minimizer over some $\mathcal{H} \subset \mathcal{H}_{all}$ as

$$f^*_{\mathcal{H}} = \arg\min_{f \in \mathcal{H}} \mathcal{R}(f)$$

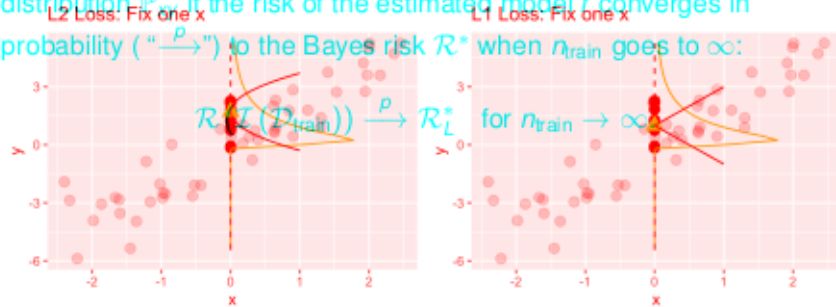
OPTIMAL POINT-WISE PREDICTIONS

To derive the risk minimizer, observe that by law of total expectation

$$\mathcal{R}(f) = \mathbb{E}_{xy}\left[L(y, f(\mathbf{x}))\right] = \mathbb{E}_{x}\left[\mathbb{E}_{y|x}\left[L(y, f(\mathbf{x}))\,|\,\mathbf{x}\right]\right].$$

- We can choose $f(\mathbf{x})$ as we want (unrestricted hypothesis space, no assumed functional form)
- Hence, for a fixed value $\mathbf{x} \in \mathcal{X}$ we can select **any** value c we want to predict. So we construct the **point-wise optimizer**

$$f^*(\tilde{\mathbf{x}}) = \arg\min_c \mathbb{E}_{y|x}\left[L(y, c)\,|\,\mathbf{x} = \tilde{\mathbf{x}}\right]$$



THEORETICAL AND EMPIRICAL RISK / 2

The risk minimizer is mainly a theoretical tool:

- In practice we need to restrict the hypothesis space $\mathcal{H}$ such that we can efficiently search over it.
- In practice we (usually) do not know $\mathbb{P}_{xy}$. Instead of $\mathcal{R}(f)$, we are optimizing the empirical risk

$$\hat{f}_{\mathcal{H}} = \arg\min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f) = \arg\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

Note that according to the **law of large numbers** (LLN), the empirical risk converges to the true risk (but beware of overfitting!):

$$\hat{\mathcal{R}}_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right) \xrightarrow{n \to \infty} \mathcal{R}(f).$$

ESTIMATION AND APPROXIMATION ERROR

Goal of learning: Train a model $\hat{f}_{\mathcal{H}}$ for which the true risk $\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right)$ is close to the Bayes risk $\mathcal{R}^*$. In other words, we want the **Bayes regret** or **excess risk**

$$\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right) - \mathcal{R}^*$$

to be as low as possible.

The Bayes regret can be decomposed as follows:

$$\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right) - \mathcal{R}^* = \underbrace{\left[\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}(f)\right]}_{\text{estimation error}} + \underbrace{\left[\inf_{f \in \mathcal{H}} \mathcal{R}(f) - \mathcal{R}^*\right]}_{\text{approximation error}}$$

$$= \left[\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}})\right] + \left[\mathcal{R}(f^*_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}_{all}})\right]$$